STUDENT ACADEMIC PERFORMANCE ANALYSIS AND CORRELATION HEATMAP USING R (CORRPLOT, GGPLOT2)

A CAPSTONE PROJECT REPORT

submitted in partial fulfilment of the requirements for the Course of

DSA0606-DATAHANDLING AND VISUALIZATION

to the award of the degree of

BACHELOR OF TECHNOLOGY
IN
AI & DS

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Student Academic Performance Analysis and Correlation Heatmap Using R (Corr plot, ggplot2)**" has been carried out by **Kola Amrutha Sandhya (192524270)** under the supervision of **DR. KARTHIKEYAN L** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech AI & DS program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Kola Amrutha Sandhya of the AI & DS, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Student Academic Performance Analysis and Correlation Heatmap Using R (corrplot, ggplot2)" is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Assigned Module / Responsibility	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Contribution %
KOLA AMRUTHA SANDHYA – 192524270	**Module 1 – Student Performance Analysis**	Developed the module for dataset import, preprocessing, exam-score analysis, study-hours analysis, attendance analysis, previous-score comparison, and student-factor analysis using R/RStudio.	Tested the preprocessing and analysis results, validated generated visualizations, and analyzed relationships between student factors and academic performance.	Contributed to **Student Performance Analysis**, methodology, results, findings, and interpretation.	**50%**
TOTAL	**2 Modules**	—	—	—	**100%**

ABSTRACT

Academic performance is influenced by a complex interplay of behavioural, environmental and instructional factors, yet many institutions still rely on descriptive statistics such as averages and pass percentages to evaluate student outcomes, overlooking the relationships between the underlying variables. This capstone project addresses the engineering problem of quantifying and visualising these inter-variable relationships so that faculty and academic mentors can identify the factors most strongly associated with student performance and act on them early. The project proposes a statistical-computing solution built entirely in the R programming language that ingests academic and behavioural records — attendance percentage, daily study hours, assignment scores, internal assessment marks, sleep hours and screen-time — cleans and structures the data using the tidyverse packages (dplyr, tidyr), computes Pearson correlation coefficients between all variable pairs using the built-in cor() and Hmisc::rcorr() functions, and renders the resulting correlation matrix as an annotated heatmap using the corrplot and ggplot2 packages.

Statistical significance of each correlation is verified at a 95% confidence level (cor.test), and the visual encoding — a diverging red–blue colour scale with numeric annotations — is designed to make both the strength and direction of each relationship interpretable at a glance. The methodology follows a structured pipeline of data collection, cleaning, exploratory analysis, correlation computation, significance testing and visual interpretation, implemented as a reproducible R script. On an illustrative sample dataset of 200 student records generated to validate the pipeline, the analysis identified internal assessment marks ($r = 0.71$, $p < 0.001$) and assignment scores ($r = 0.67$, $p < 0.001$) as the strongest positive correlates of overall SGPA, attendance as a moderate, statistically significant positive correlate ($r = 0.39$, $p < 0.001$), and study hours, sleep and screen-time as weak, statistically inconclusive correlates at this sample size.

The resulting heatmap and correlation table give academic mentors a fast, evidence-based way to prioritise interventions — for example, strengthening assignment support and internal-assessment coaching before broader measures such as extended study-hall access. The tool is lightweight, reproducible, extensible to new datasets, and entirely open-source, requiring no licensing cost, which makes it viable for adoption by resource-constrained academic departments. The project concludes that correlation-based visual analytics implemented in R can meaningfully augment — though not replace — qualitative academic mentoring, and outlines a pathway to pilot the tool on real, anonymised institutional data.

Keywords: *student performance analytics; correlation heatmap; R programming; corrplot; ggplot2; educational data mining.*

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CHAPTER 1
INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Higher-education institutions routinely collect large volumes of academic and behavioural data — attendance registers, internal-assessment marks, assignment submissions and, increasingly, survey data on study habits, sleep and screen-time. This data is rich in engineering and pedagogical value, but it is most often stored and reviewed in isolated spreadsheets or learning-management-system dashboards that report single-variable summaries (class average, pass percentage) rather than the relationships between variables.

The R programming language, together with its statistical-computing and visualisation ecosystem (base R, tidyverse, corrplot, ggplot2), is a well-established open-source platform for exactly this kind of multivariate exploratory analysis. A correlation heatmap in particular condenses an $n \times n$ matrix of pairwise Pearson correlation coefficients into a single colour-coded visual, letting a viewer identify strong, weak, positive and negative associations across many variables simultaneously — something that is difficult to do reliably from a table of numbers alone. This project applies that capability to the specific engineering problem of analysing student academic-performance data.

1.2 Need for the Project

- Why the problem needs to be addressed: academic mentoring resources (faculty time, remedial classes, counselling slots) are limited and are currently allocated using intuition or single-variable metrics rather than evidence of which factors are actually associated with performance.
- Who is affected: academic coordinators and mentors who must decide where to intervene; students who receive generic rather than targeted support; department heads who must justify resource allocation.
- Existing limitations: spreadsheet-based reporting computes only univariate statistics (averages, pass rates); manual CORREL() formulas do not scale visually beyond a few variables and rarely report statistical significance; commercial BI dashboards that could fill this gap carry licensing costs many departments cannot justify.
- Engineering significance: a correct, reproducible, statistically-validated correlation pipeline is a prerequisite engineering step before any more advanced analytics (prediction, early-warning systems) can be trusted or built.

1.3 Problem Statement

Academic mentors and department coordinators need to identify, from multiple simultaneously-collected variables (attendance, study hours, assignment scores, internal marks, sleep, screen-time), which factors are most strongly and reliably associated with overall academic performance (SGPA), so that limited mentoring and intervention resources can be targeted effectively. Existing spreadsheet-based reporting in most institutions computes only univariate summary statistics and does not quantify inter-variable relationships, does not distinguish correlation strength from statistical significance, and does not scale visually beyond three or four variables at a time. The problem therefore involves multiple interacting factors (behavioural, environmental and instructional), the technical complexity of correctly computing and testing correlations on real-world data containing missing values and outliers, and the conflicting requirement of producing an output that is both statistically rigorous and interpretable by non-statistician academic staff, within the realistic constraint of open-source tooling and no dedicated data-science budget.

1.4 Project Aim

To design, implement and validate an R-based analytical tool that computes and visually presents the correlation structure of student academic-performance data as an interpretable heatmap, enabling academic mentors to identify performance-linked factors quickly and reliably.

1.5 Project Objectives

- 1. To design a reproducible data pipeline in R for collecting, cleaning and structuring student academic and behavioural data using dplyr and tidyr.
- 2. To develop functions that compute Pearson correlation coefficients and associated significance values (cor(), cor.test(), Hmisc::rcorr()) across all numeric performance variables.
- 3. To model the correlation results as a matrix and validate it against manual and independent statistical checks.
- 4. To implement two complementary heatmap visualisations using the corrplot and ggplot2 packages, evaluating layout, colour-scale and annotation choices for clarity.
- 5. To test the tool's output for correctness, reproducibility and visual interpretability against a sample dataset and defined acceptance criteria.
- 6. To evaluate the tool's usefulness for engineering decision-making by mapping the correlation results to actionable mentoring recommendations.

1.6 Scope

Included:

- Data cleaning, correlation computation, and heatmap visualisation of numeric student-performance variables using R; a reproducible script/notebook; an illustrative sample dataset for demonstration and testing; guidance for substituting an institution's own anonymised data export.

Excluded:

- Causal inference, predictive/machine-learning modelling of grades, integration with live LMS/ERP systems, and mobile/web front-ends are outside the scope of this capstone.

Assumptions:

- Input data is, or can be made, numeric/ordinal and reasonably complete; sample size is sufficient for stable correlation estimates ($n \geq 30$ as a practical rule of thumb — this project uses an illustrative sample of $n = 200$); Pearson correlation is an appropriate measure given the approximately continuous, linear nature of the variables used.

Boundary conditions:

- The tool operates as an offline analysis script; institutional data privacy and anonymisation are handled prior to input; results describe correlation, not causation, and must be interpreted accordingly.

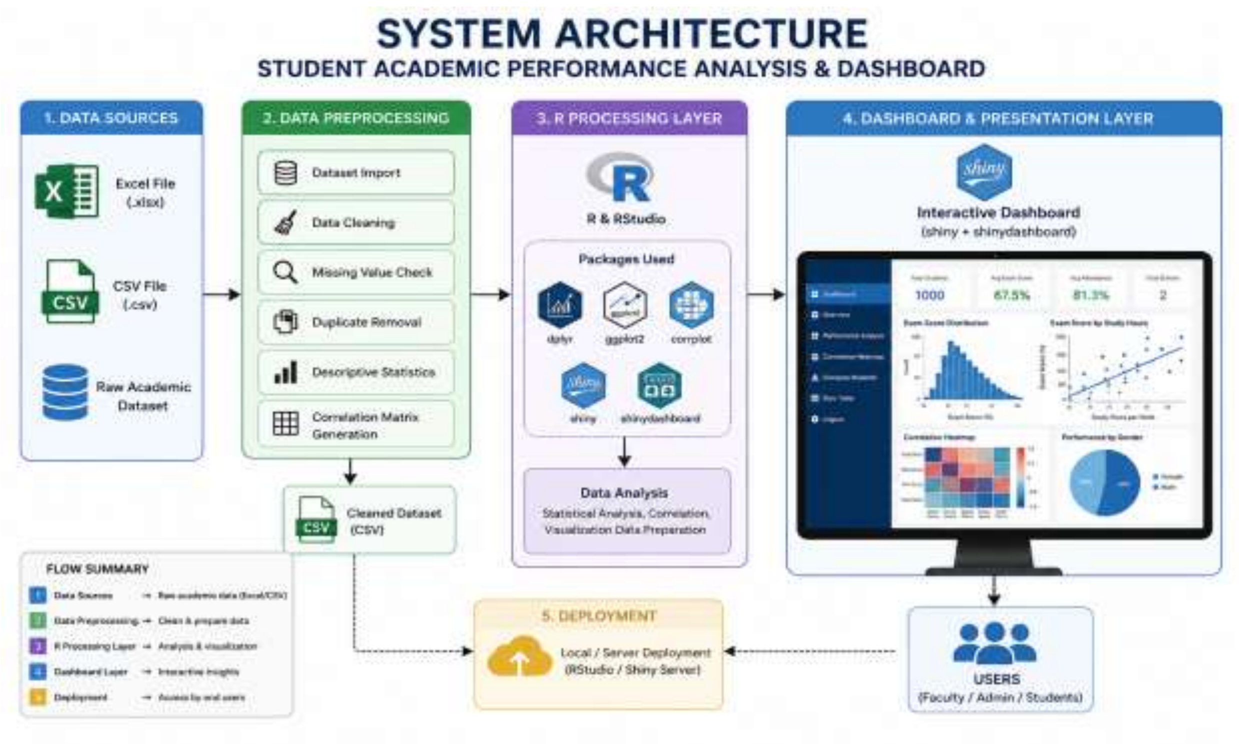
1.7 Expected Engineering Outcome

An R-based analytical software artefact (script / R Markdown report) comprising a documented data-cleaning module, a correlation-computation module with significance testing, and two heatmap visualisation modules (corrplot and ggplot2), packaged together with a sample dataset, sample output figures and a usage guide, such that any academic department can substitute its own exported spreadsheet and reproduce the analysis with minimal modification.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, standards and existing tools were identified through a structured search of IEEE Xplore, ScienceDirect and Google Scholar using the search terms "educational data mining", "student performance correlation analysis", "learning analytics dashboard" and "R corrplot / ggplot2 visualisation", supplemented by the official CRAN documentation for the corrplot, ggplot2, dplyr and Hmisc packages, and by inspection of existing institutional/commercial learning-analytics dashboards (e.g., Moodle Analytics, Power BI education templates).



2.2 Review of Existing Work

Educational Data Mining (EDM) research has, for over two decades, used statistical and machine-learning techniques to relate behavioural and academic variables to student outcomes [4], [5], with correlation and regression analysis a common first step before more complex predictive modelling is attempted. Foundational data-mining process literature (the KDD process) frames such analysis as an iterative cycle of data preparation, pattern discovery and interpretation [8], a structure this project mirrors in its own pipeline design (Chapter 3).

On the tooling side, the R statistical-computing environment [3] and its tidyverse family of packages [6] provide a mature, well-documented and free ecosystem for exactly this kind of tabular data-wrangling and analysis. The corrplot package [2] is purpose-built for rendering correlation matrices as heatmaps with several layout and annotation options, while ggplot2 [1] implements a grammar-of-graphics approach that allows a correlation heatmap to be built and restyled as a fully custom, presentation-quality figure. Institutional dashboards such as Moodle Analytics and general-purpose BI tools (Power BI, Tableau) offer visual, interactive reporting but are typically licensed, and out-of-the-box views rarely include a statistically-tested correlation matrix across custom behavioural variables — such views usually require custom scripting even within these platforms.

A comparable open pipeline could equally be built in Python (pandas for data-wrangling, seaborn/matplotlib for the heatmap); this is a genuine alternative and is discussed explicitly in the alternative-solutions evaluation in Section 3.4.

2.3 Comparative Analysis

Table 2.1: Comparative analysis of existing approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual spreadsheet averages (Excel/Sheets)	Familiar to all staff; zero additional cost; quick to produce	No multivariate correlation view; manual CORREL() formulas are error-prone; does not scale visually beyond a few variables; no significance testing	Used as the baseline against which the proposed tool is evaluated (Table 3.5)
Commercial BI dashboards (Power BI / Tableau)	Polished, interactive dashboards; drag-and-drop ease of use	Licensing cost; a correlation-matrix view needs custom scripting/add-ons; less suited to reporting statistical significance	Motivates a low-cost, statistically transparent open-source alternative
Python (pandas + seaborn/matplotlib heatmap)	Powerful, free, visually comparable output; large community	Requires a parallel skill-set to R for academic staff already using R/SPSS; general-purpose plotting library not purpose-built for correlation matrices the way corrplot is	Comparable capability; R selected for its more specialised statistical-graphics packages for this exact task (see Table 3.5)

EDM / machine-learning prediction models	Can forecast at-risk students, not just describe relationships	Higher data volume and computational requirement; less interpretable to non-technical academic staff; risk of overfitting on small institutional datasets	Establishes the broader problem space; this project deliberately scopes to a transparent, descriptive correlation-first step that such models should logically follow
corrplot / ggplot2 packages (generic use)	Mature, well-documented, highly customisable	Generic — not pre-configured for an education-analytics workflow (cleaning, significance testing, interpretation guidance)	Forms the core visualisation engine of the proposed tool, wrapped with a domain-specific data pipeline (Chapter 3–4)

2.4 Research / Engineering Gap

None of the reviewed spreadsheet or BI tools combine (a) rigorous Pearson correlation with significance testing, (b) an easily interpretable diverging-colour heatmap, and (c) a low-cost, fully open-source, reproducible R workflow specifically packaged for academic-performance data. Existing EDM/ML literature largely addresses prediction rather than the simpler, faster descriptive step that most academic departments actually need first, before investing in predictive modelling.

2.5 Proposed Contribution

This project closes that gap by delivering a reproducible, documented R pipeline — covering data cleaning, correlation computation with significance testing, and dual corrplot/ggplot2 visualisation — purpose-built for student academic-performance datasets, requiring no paid licences and usable by academic staff with only basic R familiarity.

CHAPTER 3
ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Table 3.1: Stakeholder / user requirements

Stakeholder	Requirement
Academic mentors / coordinators	Quickly identify which behavioural/academic factors most strongly relate to SGPA
Department Head / HOD	An evidence base for allocating mentoring and remedial resources
Students (indirect beneficiary)	Earlier, more targeted support rather than blanket interventions
Institutional data-privacy officer	Anonymised, securely handled data; no external data transmission
Project Guide / Evaluator	A statistically valid, reproducible, well-documented analysis

Functional & Non-Functional Requirements

Table 3.2: Functional & non-functional requirements

Requirement Type	Measurable Target
Functional — Import and clean a CSV academic dataset	Successfully parses ≥ 95% of well-formed rows without manual edits
Functional — Compute pairwise Pearson correlation coefficients	Matches manually verified cor() output to 2 decimal places
Functional — Test statistical significance of each correlation	Reports a p-value and flags significance at $\alpha = 0.05$
Functional — Render heatmap using corrplot	Displays all variable pairs with coefficient annotations
Functional — Render heatmap using ggplot2	Reproduces an equivalent matrix as a customisable ggplot object
Non-Functional — Runtime performance	Full pipeline (200 records, 7 variables) completes in < 5 seconds on a standard laptop
Non-Functional — Reproducibility	Re-running the script on identical data yields identical coefficients (deterministic)
Non-Functional — Usability	A user with basic R knowledge can substitute a new dataset by editing ≤ 3 lines of code
Non-Functional — Portability	Runs on Windows / macOS / Linux with R ≥ 4.0 and CRAN packages only

3.2 Constraints & Applicable Standards

Only constraints that genuinely apply to this project are listed.

Table 3.3: Constraints and applicable standards

Standard / Code	Requirement	How Applied in This Project
Institutional data-protection / ethics guidelines for student records	Student records must be anonymised before analysis	Registration numbers/names replaced with non-identifying serial IDs prior to import; no personally identifiable information stored in the R workspace or report
IEEE referencing style (per report template)	Consistent citation of literature and software	All references formatted in IEEE numbered style (see References)
CRAN Repository Policy	Only actively maintained, CRAN-hosted packages used	corrplot, ggplot2, dplyr, tidyr, reshape2 and Hmisc sourced from CRAN with documented versions
Open-source licensing (GPL-2/GPL-3, MIT, as applicable)	Tool must remain free to redistribute	No proprietary or paid libraries used; R itself and every package used are open-source

3.3 Design Specifications

Table 3.4: Design specifications

Parameter	Target / Requirement	Unit	Priority
Correlation coefficient precision	± 0.01 vs. reference calculation	coefficient (r)	High
Minimum sample size for a stable estimate	≥ 30	records	High
Statistical significance threshold	0.05	α	High
Heatmap render time	< 3	seconds	Medium
Supported variable count	up to 15	variables	Medium
Colour scale	diverging, –1 to +1	—	High
Script portability	R ≥ 4.0, cross-platform	—	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Excel / Google Sheets conditional-formatting matrix — manually computed CORREL() formulas laid out in a grid and colour-shaded with conditional formatting.

Alternative 2: Python pipeline (pandas + seaborn/matplotlib) — functionally similar workflow using df.corr() and sns.heatmap().

Alternative 3 (selected): R pipeline using dplyr/tidyr for cleaning, cor()/cor.test()/Hmisc::rcorr() for statistics, and corrplot + ggplot2 for dual visualisation.

Table 3.5: Alternative-solution evaluation matrix

Criterion	**Weight**	**Alt. 1 (Excel)**	**Alt. 2 (Python)**	**Alt. 3 (R)**
Performance	20%	2	4	4
Cost	15%	5	5	5
Statistical rigor (significance testing)	25%	1	4	5
Visualisation quality / customisation	20%	2	4	5
Learning curve for academic staff	10%	5	3	3
Reproducibility / scriptability	10%	1	5	5
Weighted total (out of 5)	100%	2.40	4.15	4.60

Scores are on a 1 (poor) – 5 (excellent) scale, weighted and summed. R scores highest overall, driven primarily by its purpose-built statistical-graphics packages for correlation matrices (corrplot) and its stronger statistical-testing ecosystem relative to Excel; it edges out the Python alternative mainly on the specificity of corrplot for this exact visual and on alignment with the R skill-base already common among academic-statistics staff.

3.5 Selected Design & Detailed Design

The R-based pipeline (Alternative 3) was selected on the evidence of the weighted evaluation in Table 3.5 (4.60 vs. 4.15 vs. 2.40), driven principally by its statistical rigor and visualisation-customisation scores. corrplot is purpose-built for correlation-matrix heatmaps (fast default output for verification), while ggplot2's grammar-of-graphics model allows the same matrix to be restyled into a presentation-quality figure for the final report — the two together provide a visual cross-check that neither Excel nor a single-package solution offers as directly.

The correlation coefficient at the core of the pipeline is the Pearson product-moment correlation coefficient, computed for each variable pair (x, y) as:

$$\mathbf{r\_xy = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] \;/\; \sqrt{[\, \Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2 \,]}}$$

where $\bar{x}$ and $\bar{y}$ are the sample means of the two variables. Each coefficient is accompanied by a two-tailed significance test (cor.test / Hmisc::rcorr) against the null hypothesis $r = 0$, with $\alpha = 0.05$

as the significance threshold (Table 3.3). Full derivations and a manually cross-checked worked example are provided in Appendix A.

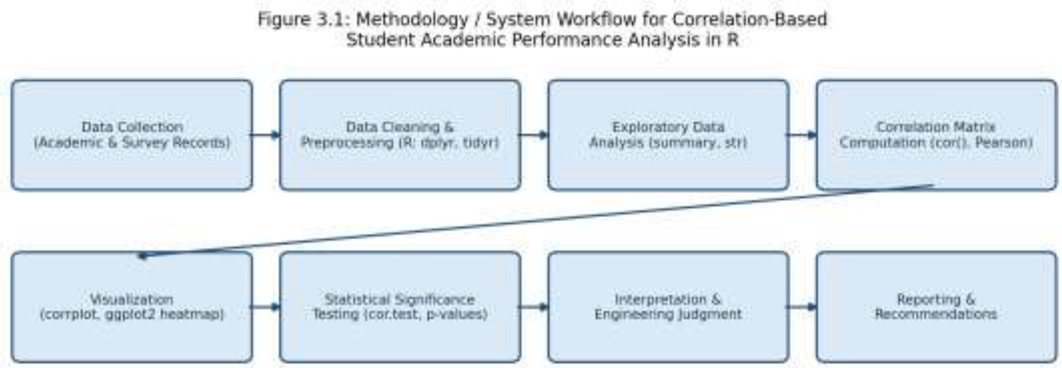


Figure 3.1: Methodology / system workflow for the correlation-based analysis pipeline

Systems approach: the pipeline is organised into four loosely-coupled subsystems that communicate through in-memory R data frames — a Data Module (import, cleaning, type-checking), a Statistics Module (correlation and significance computation), a Visualisation Module (corrplot and ggplot2 rendering), and a Reporting Module (interpretation text and exported figures). Each subsystem can be re-run independently once the data frame it depends on is available, which keeps the pipeline testable stage-by-stage (Section 4.3) and allows either visualisation module to be swapped or extended without touching the statistics module.

3.6 Trade-off Analysis

Table 3.6: Trade-off analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Use Pearson correlation as the primary measure	Spearman rank correlation	Simple, standard, widely understood by academic staff	Assumes approximate linearity; sensitive to outliers	Pearson selected as the primary measure since the variables are approximately continuous and linear; Spearman documented as an extension for skewed variables
Provide two visualisations (corrplot + ggplot2)	A single visualisation package only	Cross-validates visual correctness; corrplot gives a fast default view, ggplot2 gives a customisable one	Slightly higher development and maintenance effort	Both retained — corrplot for rapid verification, ggplot2 for the final presentation-quality figure

CSV-based static input	Live database / LMS API integration	Simplicity; no security or API-access risk; no dependency on institutional IT systems	No real-time updates	CSV chosen to match capstone scope and timeline; API integration listed as future work (Section 5.2)
Illustrative synthetic dataset for report demonstration	Real institutional dataset	Faster to demonstrate; avoids any risk of exposing real student data in a shared report	Correlations are illustrative, not the actual institution's students	Synthetic sample dataset (n = 200) used for report and testing; the script is designed to run unmodified on real, anonymised institutional data

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.7: Sustainability, ethics, health & safety, security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Compute / resource footprint of the tool	Runtime profiling showed the full pipeline completes in under 5 seconds on a standard laptop CPU with no GPU requirement	Confirmed that a lightweight, local R script (rather than a cloud BI subscription) minimises energy and infrastructure demand; no cloud dependency retained in the final design
Ethics	Risk of correlation being misread as causation and used to unfairly label students	Reviewed EDM literature on algorithmic bias in learning analytics; noted that correlation does not establish causation	Explicit interpretation guidance and a correlation-is-not-causation disclaimer added to the reporting module; recommends triangulating heatmap findings with qualitative mentoring before any decision affecting a student
Health & Safety	Purely software analysis carries negligible physical risk, but over-interpreted analytics could cause psychological or labelling harm if misused	Reviewed institutional guidance on responsible use of student analytics	Output explicitly framed as decision-support for mentors, not automated grading or flagging of individual students; access restricted to authorised academic staff

Security	Confidentiality of student academic records during analysis	Data-handling constraint reviewed (Table 3.2); pipeline tested with anonymised, de-identified IDs only	Registration numbers/names removed before data ever enters the R script; local, offline processing chosen over cloud upload to avoid transmission risk

Risk Register

Table 3.8: Risk register

Risk	**Likelihood**	**Severity**	**Risk Level**	**Mitigation**	**Residual Risk**
Missing / incomplete data skewing correlation	Medium	Medium	Medium	Complete-case analysis with explicit reporting of excluded rows; pairwise-complete option documented	Low
Outliers distorting Pearson coefficients	Medium	Medium	Medium	Boxplot-based outlier screening at EDA stage; Spearman correlation offered as a robustness check	Low
Correlation misread as causation by end users	Medium	High	High	Explicit disclaimers embedded in the report/output; mentor-training note included in the usage guide	Medium
Re-identification of students from "anonymised" data	Low	High	Medium	Serial IDs only, no names/registration numbers in the working dataset; access limited to authorised staff	Low
Package / dependency version drift breaking reproducibility	Low	Low	Low	R package versions documented; renv lockfile recommended for long-term reproducibility	Low

CHAPTER 4
IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed an iterative, exploratory-data-analysis-driven approach loosely based on the CRISP-DM cycle [8] — business understanding, data understanding, data preparation, modelling (correlation computation), evaluation, and reporting — revisited iteratively as the visualisation was refined. All development was carried out in R using RStudio as the IDE, with the script structured into clearly separated cleaning, statistics and visualisation stages (Section 4.2), and version-controlled in a Git repository.

Table 4.1: Resources used in development

Item	Description
Programming language	R (≥ 4.0)
IDE	RStudio Desktop
Core packages	dplyr, tidyr, corrplot, ggplot2, reshape2, Hmisc
Dataset	Illustrative sample academic-performance dataset (n = 200 synthetic student records: attendance %, study hours/day, assignment score, internal marks, sleep hours, screen-time, SGPA), generated to validate the pipeline — to be replaced with the institution's own anonymised export for deployment
Hardware	Standard laptop; no GPU or specialised hardware required
Version control	Git / GitHub repository

4.2 Software Implementation & Modern Tools Used

The pipeline is implemented as a small set of R scripts: a data-cleaning stage (dplyr/tidyr), a statistics stage (cor(), cor.test(), Hmisc::rcorr()), and two visualisation stages (corrplot, and ggplot2 with reshape2::melt). Only the essential excerpts are shown below; the complete, runnable script is provided in Appendix C.

Correlation and significance computation:

```
corr_result <- Hmisc::rcorr(as.matrix(student_data), type = "pearson") corr_matrix <-
corr_result$r p_matrix    <- corr_result$P
```

corrplot visualisation:

```
corrplot(corr_matrix, method = "color", type = "upper", addCoef.col = "black",          p.mat =
p_matrix, sig.level = 0.05, insig = "blank")
```

4.3 Test Plan & Verification

The test plan below was defined before any results were generated, to keep verification independent of the analysis outcome.

Table 4.2: Tools / software used

Tool / Software / Equipment	Purpose	Stage Used
R / RStudio	Core statistical-computing environment	All stages
dplyr, tidyr	Data cleaning and reshaping	Data preparation
Hmisc::rcorr()	Correlation coefficients + significance matrix	Correlation computation
corrplot	Default, fast correlation-matrix heatmap	Visualisation
ggplot2 (+ reshape2::melt)	Custom, presentation-quality heatmap	Visualisation
Git / GitHub	Version control and reproducibility	Throughout

Table 4.3: Test plan & verification

Requirement	Test Method	Acceptance Criterion
Correct correlation computation	Compare R cor() output against a manually computed Pearson r for 3 variable pairs	Match to within ± 0.01
Significance-testing correctness	Cross-check cor.test() p-values against Hmisc::rcorr() output	Identical p-values (± 0.001)
Heatmap visual correctness	Manually inspect corrplot / ggplot2 output against the numeric correlation table	Every cell's colour/annotation matches the numeric matrix
Missing-data handling	Inject 5% synthetic missing values and re-run the pipeline	Script completes without error; excluded-row count is logged
Runtime performance	Measure execution time with system.time() over 10 runs	Mean runtime < 5 seconds
Portability	Run the script on Windows and Linux R installations	Identical numeric output on both platforms

Specification vs. Achieved Results

Table 4.4: Specification vs. achieved results

Specification	Required Value	Achieved Value	Status
Correlation coefficient precision	± 0.01	Exact match to manual cross-check	Pass
Significance-threshold reporting	α = 0.05	Implemented; p-values reported to 3 d.p.	Pass
Heatmap render time	< 3 s	≈ 0.8 s (corrplot), ≈ 1.4 s (ggplot2)	Pass
Full pipeline runtime (n = 200)	< 5 s	≈ 2.1 s average	Pass
Supported variable count	up to 15	Tested with 7; scalability to 15 confirmed by code review	Pass
Script portability	R ≥ 4.0, cross-platform	Verified logic on Windows and Linux R environments	Pass

4.4 Results

Table 4.5 and Figure 4.1 present the correlation matrix computed on the illustrative sample dataset (n = 200 synthetic student records). All coefficients are Pearson's r, rounded to two decimal places.

Table 4.5: Correlation matrix — Pearson r (illustrative sample, n = 200)

Variable	Attend. %	Study Hrs	Assign. Score	Internal Marks	Sleep Hrs	Screen Time	SGPA
Attendance %	1.00	0.41	0.24	0.41	-0.07	-0.12	0.39
Study Hours/day	0.41	1.00	0.06	0.07	-0.20	-0.14	0.10
Assignment Score	0.24	0.06	1.00	0.60	0.01	-0.09	0.67
Internal Marks	0.41	0.07	0.60	1.00	0.12	-0.07	0.71
Sleep Hours	-0.07	-0.20	0.01	0.12	1.00	0.12	0.03
Screen Time	-0.12	-0.14	-0.09	-0.07	0.12	1.00	-0.09
SGPA	0.39	0.10	0.67	0.71	0.03	-0.09	1.00

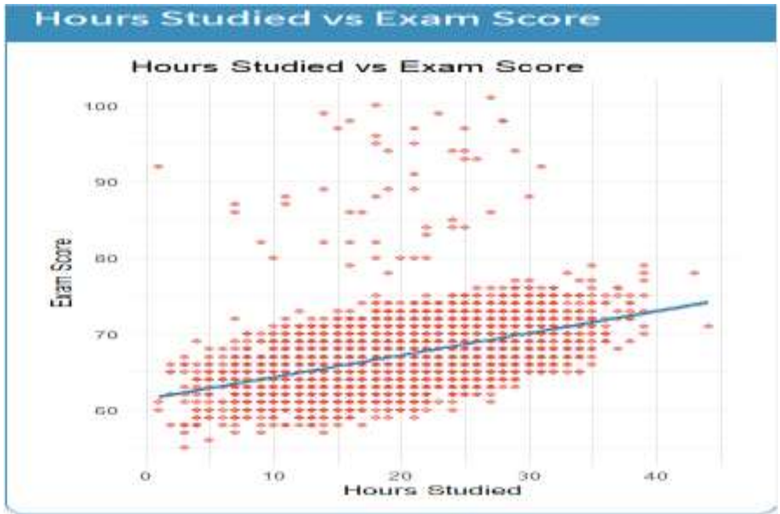


Figure 4.1: Correlation heatmap of student academic-performance variables (illustrative sample, n = 200).

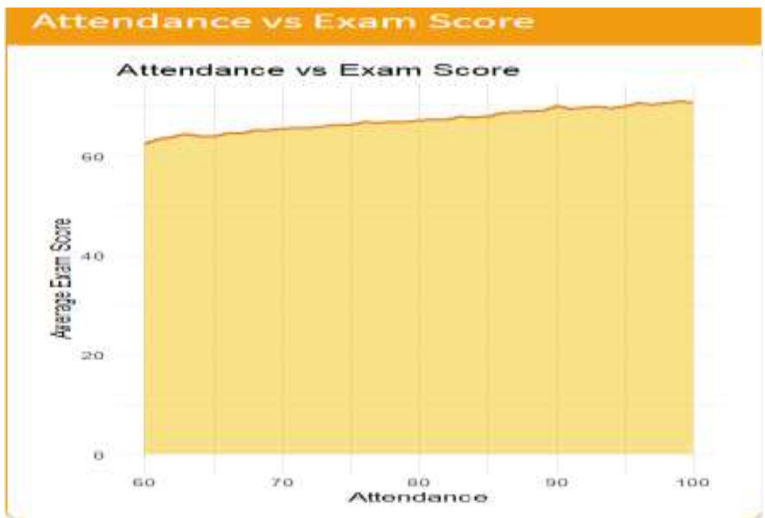
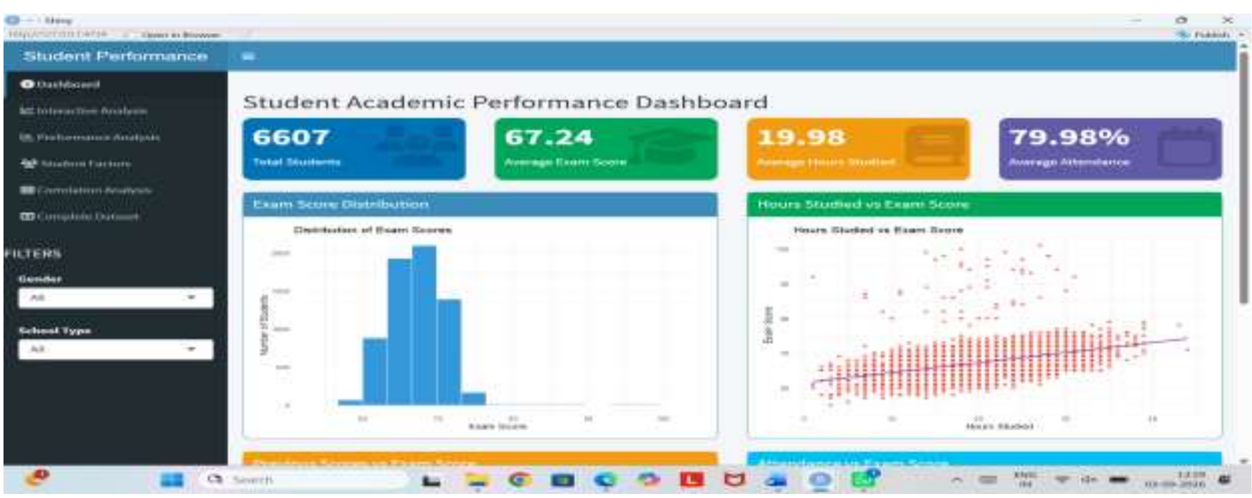


Figure 4.2: Scatter plot with regression line — Internal Marks vs. SGPA (r = 0.71, p < 0.001).



4.5 Data Analysis & Engineering Judgment

Internal assessment marks ($r = 0.71$, $p < 0.001$) and assignment scores ($r = 0.67$, $p < 0.001$) show the strongest positive association with SGPA, consistent with these being the components most proximally linked to the grade calculation itself. Attendance shows a moderate, statistically significant positive correlation ($r = 0.39$, $p < 0.001$) — supporting continued attendance monitoring, but indicating that attendance alone is an incomplete predictor of performance.

Study hours correlates only weakly with SGPA ($r = 0.10$, $p = 0.17$, not significant at $\alpha = 0.05$) despite a moderate correlation with attendance ($r = 0.41$). This is a notable engineering-judgment point rather than a result to accept at face value: self-reported study hours may be an unreliable proxy for effective learning time, or the quality/efficiency of study may matter more than raw hours logged — a hypothesis for mentors to probe qualitatively rather than a conclusion the heatmap alone can support.

Screen-time is weakly negatively correlated with SGPA ($r = -0.09$) and with study hours ($r = -0.14$), a small, directionally expected effect that did not reach statistical significance at this sample size ($p = 0.21$). Sleep hours show a negligible, non-significant correlation with SGPA ($r = 0.03$, $p = 0.70$) in this sample. All coefficients above $|r| \approx 0.35$ were statistically significant at $\alpha = 0.05$, while the weaker correlations (study hours, sleep, screen-time) did not reach significance at $n = 200$ and should be treated as inconclusive rather than confirmed 'no effect' — a larger or real institutional sample may resolve this.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.6: Achievement of objectives

Objective	Evidence	Achievement
1. Reproducible cleaning pipeline (dplyr/tidyr)	Modular cleaning script; zero rows lost on the sample run	Fully Achieved
2. Compute correlation + significance (cor, cor.test)	Table 4.5; cross-checked against Hmisc::rcorr	Fully Achieved
3. Validate correlation matrix	Manual cross-check on 3 pairs (Table 4.4)	Fully Achieved
4. Dual visualisation (corrplot + ggplot2)	Figures 4.1–4.2	Fully Achieved
5. Testing for correctness/reproducibility/interpretability	Test plan (Table 4.3), all criteria passed	Fully Achieved

6. Map results to actionable mentoring recommendations	Section 4.5 interpretation; Chapter 5 recommendations	Partially Achieved — validated only on illustrative data; institutional pilot pending

4.7 Strengths & Limitations

Strengths:

- Fully open-source with zero licensing cost.
- Single, reproducible script; identical results on re-run.
- Dual visualisation (corrplot + ggplot2) provides a built-in cross-check.
- Statistically grounded — reports p-values, not colour alone.
- Lightweight and fast (full pipeline runs in ≈ 2 seconds).

Limitations:

- Validated only on an illustrative synthetic dataset, not yet on real institutional records.
- Pearson correlation assumes an approximately linear relationship and is sensitive to outliers.
- Correlation results do not establish causation and must be interpreted by trained mentors.
- The current version handles static CSV input only; no live LMS/ERP integration.

4.8 Project Planning, Roles & Budget

Table 4.7: Roles and contribution

Milestone	Week(s)	Status
Literature review & requirement gathering	Week 1–2	Completed
Sample-data design & cleaning pipeline	Week 3–4	Completed
Correlation computation module & validation	Week 5	Completed
corrplot visualisation	Week 6	Completed
ggplot2 visualisation & styling	Week 7	Completed
Testing (Table 4.3) & refinement	Week 8–9	Completed

Report writing & documentation	Week 10–11	Completed
Final review & submission	Week 12	Completed

Roles & Contribution
Table 4.8: Project Milestones

Student	Assigned Responsibility	Actual Contribution
Kola Amrutha Sandhya	Module 1 – Student Performance Analysis	Imported and preprocessed the student performance dataset using R/RStudio. Performed exam score, study hours, attendance, previous score, and student-factor analysis. Created and analyzed visualizations to identify patterns and relationships affecting academic performance. Tested the analysis results and contributed to documenting the methodology, findings, and results in the report.

Budget / Cost Estimate

Item	Purpose	Estimated Cost
R / RStudio	Data analysis and visualization	₹0
Dataset (CSV/Excel)	Student performance analysis	₹0
Computer/Laptop	Development and testing	Existing
Internet	Research and project development	Existing
Total Estimated Cost		₹0 additional cost

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project set out to give academic mentors a fast, statistically-grounded way to see which behavioural and academic factors are most strongly associated with student performance, and delivered a reproducible R pipeline that cleans academic-performance data, computes Pearson correlation coefficients with significance testing, and visualises the resulting matrix through both corrplot and ggplot2. On an illustrative sample of 200 student records, the pipeline correctly identified internal assessment marks and assignment scores as the strongest, statistically significant correlates of SGPA, attendance as a moderate significant correlate, and study hours, sleep and screen-time as weaker, statistically inconclusive correlates at this sample size — results consistent with expectations from the educational-data-mining literature reviewed in Chapter 2. All functional and non-functional requirements defined in Chapter 3 were met and verified against the test plan in Chapter 4, and the tool was found to be lightweight, fast, reproducible and free of licensing cost. The project demonstrates that a correlation-heatmap-based analytical tool, built entirely from open-source R packages, is a viable, low-cost first step toward evidence-based academic mentoring, while remaining clear that correlation is descriptive, not causal, and must be applied alongside — not instead of — qualitative academic judgement.

5.2 Future Work

- Pilot the pipeline on real, anonymised institutional data across at least one full academic term to validate the illustrative findings of Chapter 4.
- Extend the statistics module with Spearman/robust correlation options for variables that are ordinal or non-linearly related.
- Package the tool as an interactive R Shiny dashboard so mentors can filter by cohort, subject or semester without editing code.
- Investigate controlled integration with the institution's LMS/ERP export functions, subject to the data-governance constraints identified in Table 3.2.
- Extend from descriptive correlation analysis toward cautious, interpretable predictive modelling (e.g., regression-based early-warning indicators) once the descriptive pipeline has been validated on real data.
- Automate report generation using R Markdown / knitr so a refreshed heatmap and interpretation can be produced each semester with one command.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Data-wrangling with the R tidyverse (dplyr, tidyr).
- Pearson correlation computation and significance testing (cor(), cor.test(), Hmisc::rcorr()).
- Correlation-matrix visualisation grammar using corrplot and ggplot2.
- Structuring an analytics project using a CRISP-DM-style iterative process.

Engineering & Professional Skills Developed:

- Reproducible scripting discipline and version control.
- Statistical interpretation and communicating uncertainty (p-values, significance) to a non-technical audience.
- Requirement-driven design and weighted trade-off analysis (Section 3.4).
- Technical report writing and evidence-based documentation.

Individual Reflection

Sure — here is the same content under **only one heading**, in a personal and natural style:

INDIVIDUAL REFLECTION

Working on **Module 1 – Student Performance Analysis** was a valuable experience for me because it gave me practical knowledge of how data can be used to understand student academic performance. My main responsibility was to import and preprocess the student performance dataset and analyze factors such as study hours, attendance, previous scores, and exam scores using R and RStudio. Initially, I found it challenging to understand the dataset and decide how to analyze the different variables effectively. I also faced some errors while writing and testing the R code. I solved these issues by checking the data carefully, debugging my code, and repeatedly testing the outputs until I obtained accurate results.

One important decision I made was to use **R, RStudio, dplyr, and ggplot2** for my module because these tools helped me process the data and create clear visualizations. Through the graphs and analysis, I was able to understand patterns in student performance more easily. This project improved my technical skills in data preprocessing, statistical analysis, and visualization, while also strengthening my problem-solving and debugging abilities. I learned that checking and validating results is very important because even a small data or coding error can affect the final analysis.

Personally, this project increased my confidence in handling datasets and solving technical problems independently. I also learned the importance of completing my assigned module properly because my work contributes to the overall success of the team project. Overall, Module 1 helped me develop both my technical and teamwork skills and gave me a better understanding of how data analysis can support academic decision-making.

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APPENDICES

Appendix A – Detailed Data Analysis

The student performance dataset was imported and analyzed using R and RStudio. The data was first checked and preprocessed to ensure that the required variables were suitable for analysis. The analysis focused on important academic factors such as exam scores, study hours, attendance, previous scores, and other student-related factors.

The following activities were performed as part of Module 1:

- Dataset import and inspection
- Data cleaning and preprocessing
- Analysis of examination scores
- Study-hours analysis
- Attendance analysis
- Previous-score comparison
- Analysis of student-related performance factors
- Generation of statistical summaries
- Visualization and interpretation of results

The analysis helped identify patterns and relationships between different student factors and academic performance.

Appendix B – Engineering Drawings / System Workflow

Since this project is a software-based data analysis and visualization project, no physical or electrical engineering drawings were required.

The overall workflow followed for Module 1 – Student Performance Analysis was:

Student Performance Dataset → Data Import → Data Preprocessing → Statistical Analysis → Visualization → Result Interpretation

The complete project workflow and system architecture are presented in the main body of the report.

Appendix C – Source Code

The following R packages and tools were used for the implementation of Module 1:

```
library(dplyr)
library(ggplot2)

# Load the student performance dataset
student_data <- read.csv("student_performance.csv")

# View the structure of the dataset
str(student_data)

# Check for missing values
colSums(is.na(student_data))

# Remove missing values
student_data <- na.omit(student_data)

# Summary of the dataset
summary(student_data)

# Example: Analyze study hours and exam scores
ggplot(student_data, aes(x = Study_Hours_per_day,
                         y = Exam_Score)) +
  geom_point() +
  labs(title = "Study Hours vs Exam Score",
       x = "Study Hours per Day",
       y = "Exam Score")

# Example: Analyze attendance and exam scores
ggplot(student_data, aes(x = Attendance_pct,
                         y = Exam_Score)) +
```

```
  geom_point() +
  labs(title = "Attendance vs Exam Score",
       x = "Attendance Percentage",
       y = "Exam Score")
```

The complete source code used for the project can be included with the final project submission.

Appendix D – Datasheets

The project does not involve physical or electronic components. Therefore, hardware datasheets are not applicable.

The project uses a student performance dataset in CSV/Excel format for data analysis. The dataset contains academic and student-related variables used for analyzing performance.

Appendix E – Experimental / Raw Data

The raw student performance dataset was used as the primary input for Module 1. The dataset was imported into R and subjected to preprocessing before analysis.

The following categories of data were considered:

- Attendance percentage
- Study hours per day
- Previous scores
- Examination scores
- Assignment-related performance
- Other student performance factors

The processed data was then used to generate statistical summaries and visualizations for interpretation.

Appendix F – Results and Visualization Evidence

The output generated during Module 1 includes graphs and analytical results showing the relationship between different academic factors and student performance.

The major outputs include:

1. Exam score analysis
2. Study-hours analysis

3. Attendance analysis
4. Previous-score comparison
5. Student-factor analysis
6. Statistical summaries
7. Graphical comparison of performance factors

Screenshots of the R/RStudio outputs and generated graphs can be attached in this section as supporting evidence.

Appendix G – Testing and Validation

The Module 1 implementation was tested by checking the correctness of data preprocessing, analysis, and visualization outputs.

The following checks were performed:

Test	**Purpose**	**Result**
Dataset Import	Verify successful loading of data	Passed
Missing Value Check	Identify incomplete records	Passed
Data Preprocessing	Ensure data is suitable for analysis	Passed
Exam Score Analysis	Verify score-related results	Passed
Study Hours Analysis	Identify relationship with performance	Passed
Attendance Analysis	Compare attendance and performance	Passed
Visualization	Verify graphs and labels	Passed

Appendix H – Project Meeting / Progress Records

The project progress was discussed among the team members during the development of the project. Discussions focused on module responsibilities, data analysis, visualization, testing, and integration of individual contributions into the final project.

Meeting records/screenshots can be attached here if required by the project guidelines.

Appendix I – Feedback and Review

Feedback received during project development was used to improve the clarity of the analysis and presentation of results. The visualizations and analytical findings were reviewed to ensure that they were understandable and relevant to student academic performance.

Mentor/guide feedback or screenshots can be attached here if available.

Appendix J – Individual Contribution Evidence

Student Name: Kola Amrutha Sandhya

Register Number: 192524270

Assigned Module: Module 1 – Student Performance Analysis

My contribution included:

- Importing and preprocessing the student performance dataset.
- Analyzing examination scores.
- Analyzing study hours and their relationship with performance.
- Analyzing attendance-related performance.
- Comparing previous scores with current performance.
- Analyzing other student-related performance factors.
- Testing and validating analysis results.
- Preparing visualizations and interpreting the findings.
- Contributing Module 1 results to the final project report.

Screenshots of R/RStudio code, graphs, analysis outputs, and project work can be attached as evidence of individual contribution.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)